

CAROLINE KELLY

London, UK | Portfolio: kellyc86.github.io/caroline-kelly-portfolio
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PROFILE

UCL Business Analytics graduate with 2+ years' experience across New York, Dublin and London turning complex commercial questions into clear briefs and shipped data products. Bridges the gap between creative teams and technical engineers, having built end-to-end fan classification platforms on label data and deployed predictive models directly into active business workflows. Constantly out at gigs and showcases, combining a deep love for live music with data-driven audience insights to give artists a competitive edge.

WORK EXPERIENCE

Williams Racing (Formula One), London, UK

May 2026 - Sep 2026

Commercial Data Analytics - MSc Dissertation Partnership

- Partnered with the commercial and partnerships team to turn their sponsor-acquisition questions into a machine-learning pipeline (customer acquisition modelling, weak supervision, survival analysis) that ranks the full Salesforce prospect pool weekly and surfaces a shortlist for outreach.
- Trained models on 340,000+ historical sponsorship deals, live global news and company financials to spot early buying signals, producing a weekly target list that identified future sponsors at 8× the rate of the team's previous campaigns.
- Scored every open deal in the CRM on likelihood to close from internal engagement data (emails, meetings, hospitality, deal origin) and grouped past deals into engagement archetypes, giving the team actionable insight into which patterns of contact convert.
- Applied NLP and a language model (sentence embeddings, then Claude via the Batch API) to classify global news into buying signals such as mergers, expansion and leadership change, and to resolve ambiguous company matches.
- Delivered explainable recommendations directly into Salesforce, presented findings to senior commercial stakeholders using clear data storytelling, and handed over a production-ready model on Williams-owned infrastructure.

Northwell Health / Northwell Holdings, New York, USA

Oct 2024 - Sep 2025

Administrative Fellowship

- Acted as chief of staff to the two executives leading a \$32M inpatient unit expansion, building Python, SQL and Excel KPI dashboards for weekly reporting and presenting insight to senior clinical and non-technical stakeholders to inform business decisions.
- Conducted market research for the Northwell Holdings investment team and evaluated submissions to a \$1M+ internal innovation challenge, scoring technologies against clinical and commercial criteria and turning findings into recommendations on which to pilot and fund
- Coordinated the setup of a US clinical trial for an Irish health-tech start-up, liaising with Enterprise Ireland and other trade boards and managing requirements, timelines and stakeholder sign-off from brief to delivery.

Arup, Dublin, Ireland

Aug 2023 - Sep 2024

Strategy and Insights Consultant

- Forecast EV charging demand across Ireland for government and infrastructure clients, combining demand modelling with market and technology research to recommend where infrastructure should be sited and which charger types would future-proof the network.
- Developed universal design guidelines for EV charging for the Department of Transport, researching the needs of different user audiences, from wheelchair users to older drivers, and translating their requirements into practical design recommendations.
- Ran stakeholder workshops and tender evaluations for €2M+ zero-emission transport programmes, translating policy priorities into business cases and actionable delivery plans.

EDUCATION

University College London, London, UK

Sep 2025 - Sep 2026

MSc Business Analytics

Predictive analytics, machine learning, experimentation design, forecasting, customer insight, data engineering, business strategy and analytics.

Trinity College Dublin, Dublin, Ireland

Sep 2019 - Aug 2023

BSc Engineering (2:1, Upper Second Class Honours)

Probability & statistics, advanced mathematics, management for engineers.

TECHNICAL SKILLS

Product and stakeholders: translating business strategy into requirements and analytical briefs; taking data and AI products from discovery and prototype to launch, adoption and handover; telling this story to senior and non-technical teams

Languages and data: Python, SQL, BigQuery (GCP), Apache Spark, Parquet and Medallion pipelines, Salesforce, PostgreSQL, MongoDB, API and external data integration (Spotify, YouTube, Reddit, GDELT, Orbis)

Machine learning and analytical aptitude: predictive and ranking models (gradient boosting), survival analysis, weak supervision (Snorkel), segmentation, NLP and sentiment analysis (transformers), explainability (SHAP), forward-time validation and leakage control, MLflow

GenAI and agentic tools: LLM-based classification and entity matching, RAG and vector search, LangChain agents, prompt design and evaluation, Batch API, Claude Code, MCP, agentic tool benchmarking

Engineering practice: FastAPI (REST, WebSocket), Docker, AWS (S3, RDS, ECR), Terraform, GitHub Actions CI/CD, pytest, Git/GitHub, data lineage (OpenLineage, W3C PROV), reproducible builds and automated data-integrity checks

SELECTED PROJECTS (live demos and case studies: kellyc86.github.io/caroline-kelly-portfolio)

- **UMG Audience Intelligence Platform:** Built to answer a label question: who are an artist's fans, how do they behave across platforms, and which segments should a campaign target? Ingested public fan data from Reddit, YouTube, Spotify, Last.fm and Fandom for five artists through a Medallion pipeline (Spark, Parquet), classified 4,800+ fans into four behavioural archetypes with weak supervision, and scored sentiment with a transformer model. Served as a working product: FastAPI (REST, WebSocket, MCP) with a RAG agent over MongoDB vector search and a locally hosted Llama 3 for data privacy, deployed with Docker and Terraform on AWS, CI/CD with 57 tests, MLflow and OpenLineage lineage. Independent coursework built around a UMG-style brief.
- **Benchmarking agentic coding tools for data science:** Designed a rubric-based harness to evaluate three agentic coding tools (Claude, Codex, Antigravity) across four data science tasks on 48,895 Airbnb listings, with Git branch isolation and fixed seeds. Found that process compliance and output quality diverge: the highest-scoring tool shipped a baseline with R^2 of -0.625 without raising an error, and none of the three self-detected its failures. Practical evidence for where AI-assisted development speeds delivery and where human review remains the binding constraint.
- **Spotify genre classification:** Predicted genre for 10,000 Spotify tracks from audio features alone (test macro-F1 0.657 vs 0.018 baseline, LightGBM chosen on cross-validation). Registered three hypotheses before modelling; all three held. The twist for campaigns: pop, indie and hip-hop tracks sound so alike that the model could barely separate them, which means genre is a positioning choice rather than a property of the sound. A track that sits on the border can be pitched to the adjacent genre's larger playlists and audiences, and the audio will not give it away. The story for marketing and creative teams: use the model to find border tracks, then decide where to position them for reach.